Theorem 16: inapproximability of the optimal-transport local edit distance

Correspondence between the paper and the Lean 4/Mathlib formalisation

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport* (file `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository). Statement, equation, page and line numbers refer to this PDF; `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

The folder `GsimOT/Theorem16_LocalEditHardness` formalises the reduction of Theorem 16 (p. 12, proof in Appendix G, pp. 51–52) from Three-Partition. It gives the case $p = 1$ of Theorem 4 (2). Shared notions come from `GsimOT/Common`; they are listed at the end.

Checking. Run `bash build.sh` in the repository root (see `README.md`). The script builds the library and runs `scripts/CheckAxioms.lean`, which must report only the standard axioms `propext`, `Classical.choice`, `Quot.sound`: no `sorry` and no additional axioms.

Main statement in Lean. For an instance $(A, a_1, \dots, a_{3m})$ of Three-Partition, let F_1 be the union of paths on a_i vertices and F_2 the union of m cycles on A vertices. `GsimOT.Hardness.ThreePartition.dOTOne_le_of_isYes`: for a YES-instance with all $a_i \geq 2$, $d_1^{\mathrm{OT}}(F_1, F_2) \leq 1$. `GsimOT.Hardness.ThreePartition.two_le_dOTOne_of_not_isYes` for a NO-instance, $d_1^{\mathrm{OT}}(F_1, F_2) \geq 2$.

Correspondence

Paper	Page	Statement	Lean declaration(s) and location
Proof of Thm. 16, l. 1661	52	$L_Q(v,w) = \deg(v) + \deg(w) - 2\,Q(N(v), N(w))$ for $Q \in \mathrm{DS}$ (the fractional form of the mismatch count $(2k - \ell) + (dk - \ell)$)	`GsimOT.Hardness.nbrMass` `Theorem16_LocalEditHardness/Theorem16.lean:55` `GsimOT.Hardness.localCost_eq` `Theorem16_LocalEditHardness/Theorem16.lean:59`
Theorem 16 (YES), l. 1626–1632	12, 51	YES-instances: $d_1^{\mathrm{OT}}(F_1, F_2) \leq 1$	`GsimOT.Hardness.ThreePartition.dOTOne_le_of_isYes` `Theorem16_LocalEditHardness/Theorem16.lean:80`
Theorem 16 (NO), Eq. (44), Cases 1 and 2, l. 1633–1666	12, 52	NO-instances: $d_1^{\mathrm{OT}}(F_1, F_2) \geq 2$	`GsimOT.Hardness.ThreePartition.two_le_dOTOne_of_not_isYes` `Theorem16_LocalEditHardness/Theorem16.lean:105`
Theorem 16 (arbitrary instances), l. 1622–1624	51	Both bounds after adding 3 to each a_i and 9 to A; this is the case $p = 1$ of Theorem 4 (2)	`GsimOT.Hardness.ThreePartition.dOTOne_shift_gap` `Theorem16_LocalEditHardness/Theorem16.lean:177`

Shared notions used (from `GsimOT/Common`)

Paper	Page	Statement	Lean declaration(s) and location
Section 2, Eq. (7)	5, 14	$\|B_{1\to1}(G,H,Q)\|_{1\to1} = C_1(Q) = \max_{(v,w)\in\mathrm{supp}(Q)} L_Q(v,w)$ for $Q \in \mathrm{DS}(V,W)$; hence $d_{1\to1}^{\mathrm{OT}} = d_1^{\mathrm{OT}}$	`GsimOT.opNorm_Bpq_one_one_eq_costOne` Common/MatrixNorms.lean:150 `GsimOT.dOT_one_one_eq_dOTOne` Common/MatrixNorms.lean:164
Proofs of Thms. 15–17	50–57	Disjoint unions of paths (F_1) and of cycles; their degrees and edge counts	`GsimOT.Hardness.pathUnion` Common/PathsAndCycles.lean:56 `GsimOT.Hardness.cycleUnion` Common/PathsAndCycles.lean:226 `GsimOT.Hardness.sum_adj_pathUnion` Common/PathsAndCycles.lean:134 `GsimOT.Hardness.deg_cycleUnion` Common/PathsAndCycles.lean:238
Proof of Thm. 15	50	Three-Partition instances $(A, a_1, \ldots, a_{3m})$; the step of adding 3 to each a_i and 9 to A	`GsimOT.Hardness.ThreePartition` Common/ThreePartition.lean:26 `GsimOT.Hardness.ThreePartition.IsYes` Common/ThreePartition.lean:39 `GsimOT.Hardness.ThreePartition.shift` Common/ThreePartition.lean:206 `GsimOT.Hardness.ThreePartition.shift_isYes_iff` Common/ThreePartition.lean:223
Proofs of Thms. 15–17	50–56	A solution gives a bijection mapping each path into one component; conversely a $Q \in \mathrm{DS}$ mapping every path into a single component gives a solution (counting mass)	`GsimOT.Hardness.ThreePartition.layoutEquiv` Common/ThreePartition.lean:137 `GsimOT.Hardness.ThreePartition.isYes_of_isDS` Common/ThreePartition.lean:168
Proofs of Thms. 17, 18	54, 57	Attaching $\ell(b)$ leaves to every vertex b of a base graph	`GsimOT.Hardness.addLeaves` Common/Leaves.lean:33

Notes on the formalisation

1. Only the correctness of the reduction is formalised: for the constructed graphs, YES-instances have a small and NO-instances a large distance. The complexity side (polynomial running time, NP-hardness of Three-Partition, Håstad's bound for MaxCut) is not formalised. The results of Section 4 and their shared gadgets are in the namespace `GsimOT.Hardness`.

2. d_1^{OT} is `GsimOT.dOTOne`, the definition of Proposition 8 and Theorem 9. By Eq. (7) (`GsimOT.dOT_one_one_eq_dOTOne`) it equals the operator-norm distance $d_{1\to1}^{\mathrm{OT}}$.

The paper bounds $d_1^\bullet$ via permutations of blowups and then applies Theorem 1 (p. 52, l. 1667). The proof works directly with doubly stochastic matrices, with the same case distinction: a row of Q split over two cycles (Case 2) or adjacent vertices sent to different cycles (Case 1) gives a local cost ≥ 2, and otherwise the counting after Eq. (44) gives a solution.

3. The formalisation assumes $a_i \geq 2$ for the YES-direction (`GsimOT.Hardness.ThreePartition.dOTOne_le_of_isYes`); the NO-direction has no hypothesis on the a_i.